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Abstract

Global implementation of sustainable energy has seen the need for new powering paradigms where such traditional wind turbines are inefficient more than ever, where the wind speed is low. Research underway is about a new emerging method for energy creation by taking advantage of vortex-induced vibrations (VIV), where aerodynamic bluff bodies are utilized for the conversion of fluid-structure mechanisms into usable electrical energy. The system consists of a NACA airfoil-shaped bluff body due to desirable aerodynamic properties such as active vigorous shedding for vortices, a low value for the coefficient of drag, and maximized oscillating amplitude. A second bluff body in the cylindrical form is utilized in the wake region for the airfoil for the purpose of extracting turbulence-induced vibrations (TIV) for further improved energy capture. The two bluff bodies are mechanically attached to permanent magnetic linear induction generators (PMLIGs), which is a direct and efficient method for converting the linear oscillating motion into usable electrical energy.

Its airfoil design also includes a dynamically shiftable dynamic vortex generator for adaptive variation in oscillation direction-induced lift forces. Its efficiency also boosts system-wide system response alongside energy conversion efficiency. Because the dual-bluff-body design is expressly optimized toward low-speed operation for wind regimes slower than 5 m/s, it is a small, scalable, economically viable urban as well as offshore-based low-speed wind energy harvesting solution. Because it is a result of low-drag design characteristics, it is particularly ideal for operation on mobile vehicles like ships where motion resistance is needed for low motion resistance. The new work enriches low-speed wind energy harvesting innovation by providing an original high-efficiency system by virtue of synergistic fluid-structure interactions.

Introduction

The global need for renewable and sustainable energy sources has inspired novel techniques of harvesting energy from environmental flows, particularly at low speeds where traditional wind turbines are inefficient. Low-speed wind energy harvesting by vortex-induced vibration (VIV) is a novel technique of electricity generation in inconsistent windy areas, such as urban regions. VIV is a fluid phenomenon whereby alternating vortices are formed by interacting with a bluff body, which produce forces to cause the body to oscillate. The oscillations can be converted to electrical energy using permanent magnet linear induction generators (PMLIGs).

This project is focused on the design of a NACA airfoil-shaped bluff body for a vibration energy harvesting system. The NACA airfoil shape is utilized because of its aerodynamic properties, which produce strong vortex shedding patterns, minimal drag force, and enhance the vibration amplitude. There is a turbulent wake area with complex flow patterns that has residual energy when wind passes over the trailing edge of the main bluff body. To maximize this energy harvest, a second bluff body(cylinder) is placed in this turbulence region to extract additional energy from the wind flow, also mounted on a PMLIG. This two-bluff-body arrangement is designed to increase overall energy harvesting efficiency.

An airfoil is defined as the cross section of a body that is placed in an airstream in order to produce a useful aerodynamic force in the most efficient manner possible. It is an aerodynamic shape that moves through the air when applied. When it is applied as wing air is split into two streams. Among them, one passes above and the other passes below the wing. The wing's upper surface is so shaped that air rushing over the top speeds up and stretches out. This phenomenon produces a pressure reduction above the wing. Comparatively, air flows in a straighter line below the wing. Thus, speed and air pressure remain the same for the shape. The aerodynamic characteristic of an airfoil is mainly dependent on the flow characteristic. Because a wing, which is an airfoil,

generates lift due to its characteristic shape. Lift acts on the center of pressure at the perpendicular of relative wind flow where drag is parallel to relative wind flow which opposes the motion of airfoil. To maximize the velocity of wind on top of the airfoil, the additional vortex generator (additional geometry) is mounted on the top face of the airfoil. That enhance the overall vortex intensity beyond the trailing edge and produce much greater lift force for oscillations. The mounted vortex generator can also move to top face and bottom face by applied mechanism, this make change the lift force induction when oscillations change its moving direction.

The second bluff-body is cylindrical one that experience extreme turbulence intensity that has highly unpredictable flow patterns. Hence, when cylindrical bluff body interact with the flow Turbulence Induce Vibrations (TIV) are causes to oscillates the body with lower amplitude and higher frequency. The purpose is to implement two bluff body arrangement is the extract maximum energy from wind via VIV and TIV.

The best and efficient method for converting linear oscillatory motion to electrical current is implement the Linear Induction Generators (LIG) that frequently use in the wave energy harvesting. The most efficient and simple one is the Permanent Magnet Liner Induction Generator (PMLIG).

The system is targeted towards applications where low wind speeds (typically below 5 m/s) are a limitation for the use of conventional turbines. By utilizing VIV and a novel two bluff-body design, the project seeks to develop a compact, scalable, and economically feasible solution for renewable energy harvesting. Also, this project is reliable as renewable energy source for ships, because the design produces minimal drag force. Therefore, resistance to movement of ship produce by the design is minimal or can be ignore.

Literature Survey

Energy harvesting from low-speed wind using vortex-induced vibration (VIV) has been a vibrant research topic in recent years due to its potential for small-scale clean energy. This literature survey brings together significant research and advancements in VIV-based energy harvesting, bluff body shapes, exploiting turbulence, and linear induction generators, which forms the foundation of the proposed dual-bluff-body arrangement with a NACA airfoil profile.

Vortex-Induced Vibration for Energy Harvesting Bernitsas et al. (2008) suggested the VIVACE (Vortex-Induced Vibration Aquatic Clean Energy) system which harnesses energy from fluid flow utilizing cylindrical bluff bodies. Their study demonstrated that VIV can be utilized to effectively convert mechanical oscillations into electrical energy using linear generators, particularly in low-velocity flows (Bernitsas, M. M., et al., 2008, *Journal of Offshore Mechanics and Arctic Engineering*). The study highlighted the importance of bluff body geometry for the optimization of vortex shedding frequency and amplitude. However, their application was more directed towards aquatic environments, with some potential for application extension to atmospheric wind applications.

Bluff Body Geometry and Aerodynamic Performance: The choice of bluff body shape has a significant effect on VIV performance. Rostami and Armandei (2017) investigated various shapes of bluff bodies, including cylinders, triangles, and airfoils, for wind energy harvesting (*Renewable and Sustainable Energy Reviews*, 2017). It was found that non-circular shapes, such as airfoils, have the potential to enhance the periodicity of vortex shedding and vibration amplitude due to their aerodynamic shapes. The NACA airfoil, which is prevalent in aeronautics, was known to manage flow separation and the creation of vortices and thus was a promising candidate for VIV applications. Few studies exist, however, that address NACA airfoils specifically in VIV-based energy harvesting.

Turbulence and Wake Utilization: The wake of a bluff body is turbulent and contains residual energy that can be harvested. Kim et al. (2016) explored the placement of secondary bodies in the wake of a primary bluff body for energy harvesting from turbulent fluctuations (*Journal of Wind Engineering and Industrial Aerodynamics*, 2016). Their experiments on tandem cylinders showed that the downstream body can be exposed to increased oscillations via wake interactions, with higher energy output. This principle validates

the proposed use of a secondary bluff body in the turbulent region, though it is less explored for airfoil-shaped bodies.

Permanent Magnet Linear Induction Generators: Harvesting VIV-induced vibration as electrical energy is generally done through linear generators. The principles of permanent magnet linear induction generators (PMLIGs) were first presented by Boldea and Nasar (1997), where they emphasized their efficiency at converting linear motion into electricity (Linear Electric Actuators and Generators, 1997). Recent studies, e.g., Wang et al. (2020), have optimized PMLIG setups for low-frequency vibrations typical of VIV, with higher power densities (Energy Conversion and Management, 2020). These advances are key to the proposed system, where low-amplitude vibrations from primary and secondary bluff bodies must be efficiently converted.

Challenges of Low-Velocity Wind Energy Harvesting: Low-velocity wind is problematic due to weaker vortex shedding and reduced vibration amplitudes. Energy harvesting from low-speed flows was discussed by Zhu et al. (2019), where VIV systems are confronted with efficiency issues at wind speeds lower than 3 m/s (Applied Energy, 2019). Novel bluff body geometries and wake utilization mechanisms are solutions to improving performance, according to their research. The proposed dual-bluff-body system surmounts this via the synergistic combination of primary VIV and turbulence-induced vibrations.

Gaps and Opportunities: While significant progress has been made in VIV-based energy harvesting, few studies have explored the use of NACA airfoil-shaped bluff bodies in atmospheric wind conditions. Additionally, the integration of a secondary bluff body to harness turbulent wake energy is a relatively new concept, with limited research on its application to airfoil-based systems. The combination of these elements with PMLIGs offers a novel approach to improving energy harvesting efficiency at low wind speeds.

In conclusion, the literature brings out the potential of VIV for low-speed wind energy harvesting, with bluff body shape, wake interaction, and efficient energy conversion mechanisms as critical factors. The proposed system here builds on these observations by introducing a NACA airfoil-shaped bluff body and a secondary bluff body in the turbulent wake, coupled with PMLIGs, to achieve optimum energy extraction efficiency.

Methodology

The project "Low Velocity Wind Energy Harvesting for Power Generation" methodology is a step-by-step process to design and analyze a VIV energy harvesting system consisting of a NACA-4 airfoil-shaped bluff body and a secondary bluff body in the turbulent wake, both connected to PMLGs. The methodology consists of CFD analysis using ANSYS Fluent for choosing the best bluff body shape, vibration characteristics calculation, PMLG design, and generated power estimation. The methodology is outlined in the following major steps:

1. Bluff Body Shape Analysis and Selection

To improve the effectiveness of vortex-induced vibrations, various bluff body shapes were investigated to evaluate their ability to create strong, reliable, and regular vortex shedding at low wind speeds of 1–5 m/s. The shapes studied were circular cylinders, triangular prisms, NACA-4 airfoil shapes, and square shapes. The NACA-4 airfoil was used due to its improved ability to support regular vortex shedding, demonstrate larger lift coefficient variations, and achieve regular wake patterns compared to other geometries, especially at low Reynolds number values ($Re \approx 10^3$ – 10^4). These were investigated using the ANSYS Fluent computational fluid dynamics software, through a standardized set of methodology steps,

Geometry Creation: 2D models of each bluff body shape were created in ANSYS Design Modeler. The NACA-4 series airfoil (NACA 0012) was defined using standard four-digit parameters to ensure a streamlined profile conducive to controlled vortex shedding.

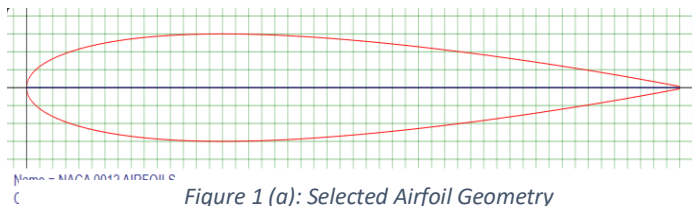


Figure 1(a): Selected Airfoil Geometry

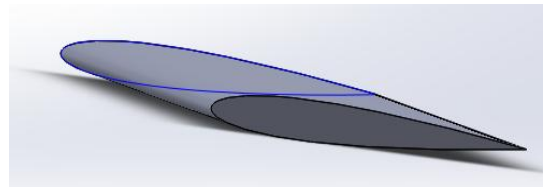


Figure 1(b): 3D model of airfoil

Meshing: The meshes were systematically generated around every bluff body, using smaller elements near the surface to efficiently capture boundary layer effects, whereas larger elements were used in the far field to keep the computational cost low. Mesh independence analyses were performed to ensure the accuracy of the results.

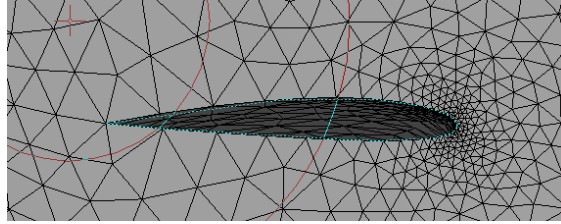


Figure 2: Meshing

CFD Simulation Setup: The simulations used a 2D incompressible flow model with the $k-\omega$ SST turbulence model to accurately capture vortex shedding and wake dynamics. Boundary conditions included a uniform inlet velocity (1–5 m/s), a pressure outlet, and no-slip walls on the bluff body surface. The domain was sized to minimize boundary effects (10 times the characteristic length of the bluff body).

Analysis of Vortex Shedding: The Strouhal number ($St = fD/U$, where f is vortex shedding frequency, D is the characteristic dimension, and U is wind velocity) was calculated for each shape to assess vortex shedding frequency. Lift and drag coefficients were analyzed to evaluate the oscillatory forces inducing VIV.

For a NACA 0012 airfoil, the Strouhal number (St) is ranging from 0.18 to 0.22 for bluff bodies or airfoils at moderate angles of attack, based on experimental data for vortex shedding in the wake. The frequency can be derived from the Strouhal number:

$$f = \frac{St \cdot U}{L}$$

The characteristic length of airfoil, $L=0.06\text{m}$. U is the stream velocity. Then frequency of vortex shedding is,

$$f = \frac{St \cdot V}{L} = \frac{0.2 \cdot V}{0.06} \text{ Hz}$$

The shedding period is:

$$T = \frac{1}{f} \text{ s}$$

Vortex shedding frequency Variation with wind velocity showing in the graph bellow,

Frequency vs Wind Velocity

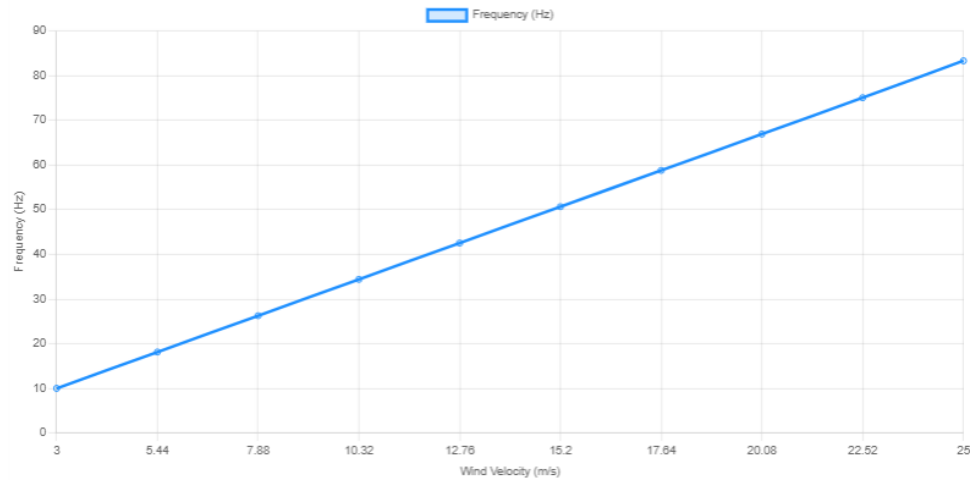


Figure 3: Vortex shedding frequency vs. wind velocity

Calculation of forces

The lift and drag forces are calculated using:

$$L = \frac{1}{2} \rho V^2 S C_L$$

$$D = \frac{1}{2} \rho V^2 S C_D$$

Where:

ρ : Air density (kg/m³)

V : Free-stream velocity (m/s)

S : Reference area (m², typically wing area for airfoils)

C_L : Lift coefficient (dimensionless, depends on airfoil shape and angle of attack)

C_D : Drag coefficient (dimensionless, depends on airfoil shape, angle of attack, and Reynolds number)

Moment is similarly defined:

$$M = \frac{1}{2} \rho V^2 S c C_M$$

Where:

c : Chord length (m, the distance from leading edge to trailing edge)

C_M : Moment coefficient (dimensionless)

The Reynolds number (Re) characterizes the flow regime:

$$Re = \frac{\rho V c}{\mu}$$

Where:

μ : Dynamic viscosity of air (kg/m·s)

c : Chord length (m)

2. Calculation of Vibration Amplitude and Frequency

Vibration characteristics for the base case NACA-4 airfoil bluff body were also determined for the purpose of outlining the VIV response driving the PMLG. The procedure included:

Dynamic Modeling: A spring-mass-damper system was employed as a representation for the bluff body subjected to periodic lift forces by virtue of vortex shedding. The equation for motion is:

$$m\ddot{y} + c\dot{y} + ky = F_L(t)$$

where (m) is the mass of the bluff body, (c) is the damping coefficient, (k) is the spring stiffness, (y) is displacement, and ($F_L(t)$) is the time varying variation for the lift force by virtue of the help by the CFD simulations. Since there are no dampers, friction and other losses to motion is consider as the dampers.

Spring Selection

To estimate the wire thickness of a spring, we can use the spring design formula related to its stiffness (spring constant, k), geometry, and material properties. The spring constant k for a helical spring is given by:

$$k = \frac{Gd^4}{8D^3N}$$

Where:

$$k = 90 \text{ N/m}$$

G = Shear modulus of the material (80 GPa)

d = Wire diameter

D = Mean coil diameter

N = Number of active coils

Since maximum vibration frequency equals to the natural frequency of spring-mass system, no damper required.

Vibration Frequency: Vortex shedding frequency output by Fluent ANSYS was utilized for VIV frequency calculation by considering lock-in conditions where system natural frequency is equivalent to shedding one ($f \approx f_n$). The natural frequency is computed by:

$$[f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}]$$

The system was also resonance for low wind speeds by manipulating (k) and (m).

Vibration Amplitude: The oscillation amplitude value was computed by numerically finding the motion equation using the data for the lift force by the CFD. Data was also verified using empirical relations for VIV like:

$$[A^* = \frac{A}{D} \approx 0.5-1.0]$$

where the characteristic length of bluff body is shown by (D) and the amplitude is shown by (A). The NACA-4 airfoil exhibited the amplitudes suitable for optimal energy transmission to the PMLG.

Secondary Bluff Body in Turbulent Wake: A second bluff body was also set inside the turbulent wake in order to gain more energy from the turbulent flow.

Wake Analysis: The flow in the wake area was also modeled using ANSYS Fluent to visualize the velocity fluctuations and the vortex structures further downstream of the primary bluff body. The wake area was seen to exist approximately 2–5 D away from the trailing edge.

Vortex shedding Location

The vortices are formed and are highly coherent in the near wake, generally within an area a few chord lengths behind the trailing edge. Maximum vortex shedding is where the vortices are highest in coherency and fullest in appearance, generally a distance within 0.5 to 2 chord lengths downstream of the trailing edge, according to experimental research as well as computation research for airfoils with low Reynolds number conditions.

The velocity of the convection of the vortices in the wake is about the free-stream velocity (3 m/s). The distance d where the first coherent vortex is located can be estimated as:

$$d \approx V \cdot t_{formation}$$

Where $t_{formation}$ is the time for vortex formation, typically on the order of 1–2 shedding periods. Using one shedding period and velocity at 3m/s:

$$d \approx V \cdot T = 3 \cdot 0.1 = 0.3 \text{ m}$$

Secondary Bluff Body Shape: A small cylinder was chosen for the second bluff body for creating VIV during the turbulent wake. Its location was optimized using modeling by CFD for achieving a maximum oscillating force induced by the wake.

Characteristics of Vibration: The vibration frequency and the secondary bluff body's amplitude were computed by the dynamic model using input forces from the turbulent wake flow data for the primary bluff body.

3. Permanent Magnet Linear Induction Generator Design (PMLG)

PMLG was thus conceptualized for the purpose of converting oscillatory motion in a straight line for bluff bodies into electricity. The system comprised:

PMLG Configuration: A permanently magnetized mover is present on the oscillating bluff body with a stator made by using copper coils. Neodymium magnets were selected for maximum magnetic flux density while coil design was optimized for maximum induced electromotive force (EMF) generation.

Electromagnetic Calculations: Induced EMF was computed according to Faraday's law:

$$\mathcal{E} = -N \frac{d\Phi}{dt}$$

where turns for the coil are represented by (N) and magnetic flux by (Phi) depending on oscillating motion for the magnet proportional to the vibration rate as well as the amplitude. Design Parameters: The PMLG was also optimized for low-frequency operation (1–10 Hz) and low-amplitude operation (1–10 cm), typical VIV for low wind speed. These design parameters are optimized either using electromagnetic simulation software such as ANSYS Maxwell and by analytical models.

The magnet's motion is sinusoidal due to VIV, with displacement $x(t)$:

$$x(t) = x_0 \sin(2\pi ft)$$

x_0 : Amplitude of oscillation .

f : Frequency of oscillation (Hz).

t : Time (in seconds).

The velocity of the magnet is the derivative of displacement:

$$v(t) = \frac{dx}{dt} = 2\pi f x_0 \cos(2\pi ft)$$

Induced EMF is:

$$EMF = -NBA2\pi f \cos(2\pi ft)$$

The electrical power generated depends on the load resistance R_{load} and the coil's internal resistance R_{coil} :

$$P = \frac{EMF^2}{R_{load} + R_{coil}}$$